Automobile Engineering

CSE - Semester 1 - Steering System - 16 August 2026

Module 2: Steering System

Camber

Camber is the inward or outward tilt of the wheel when viewed from the front of the vehicle. Positive camber means the top of the wheel tilts outwards, while negative camber means the top tilts inwards. It's measured in degrees.

Correct camber improves tire contact with the road, enhancing grip, stability, and steering response. Incorrect camber leads to uneven tire wear, reduced traction, and poor handling.

caster - king pin inclination - toe-in/out

Caster

Caster is the forward or backward tilt of the steering axis when viewed from the side of the vehicle. Positive caster means the top of the steering axis is tilted rearward, while negative caster means it's tilted forward.

Positive caster provides directional stability, helping the wheels return to the straight-ahead position after a turn. It also improves steering feel and high-speed stability, like a shopping cart wheel.

camber - king pin inclination - steering geometry

King Pin Inclination (KPI) / Steering Axis Inclination (SAI)

KPI is the inward tilt of the steering axis when viewed from the front of the vehicle. It's the angle between the steering axis and the vertical line through the wheel center.

KPI reduces steering effort, provides steering self-centering, and minimizes tire scrub during turns. It also creates a 'scrub radius' which influences steering stability and feedback.

camber - caster - scrub radius

Toe-in and Toe-out

Toe is the difference in distance between the front and rear of the tires when measured across the wheel. Toe-in means the front of the wheels are closer together than the rear. Toe-out means the front are further apart.

Toe settings compensate for forces acting on the wheels while driving, ensuring optimal tire contact and reducing tire wear. It affects straight-line stability and cornering response.

camber - caster - steering geometry

Understeering vs Oversteering

	Definition	Vehicle Behavior	Correction
Understeering	Loss of front wheel grip, vehicle turns less than intended.	Vehicle pushes wide in a turn, 'plowing' straight ahead.	Reduce throttle, lightly brake, reduce steering input.
Oversteering	Loss of rear wheel grip, vehicle turns more than intended.	Rear end slides out, vehicle wants to spin.	Counter-steer into the slide, modulate throttle.

Steering Geometry

Steering geometry refers to the precise angles and relationships of the steering and suspension components. It includes camber, caster, king pin inclination, and toe settings, all designed to work together.

Proper steering geometry is crucial for safe and predictable vehicle handling, tire longevity, and driver comfort. It ensures accurate steering response and stability at all speeds.

camber - caster - toe-in/out - Ackermann steering

Ackermann Steering Mechanism

The Ackermann steering mechanism ensures that during a turn, all wheels roll without slipping. The inner wheel turns at a sharper angle than the outer wheel, so their turning centers coincide.

> The Ackermann principle states that the axes of all wheels should intersect at a common point on the extension of the rear axle during a turn.

This mechanism minimizes tire scrub and wear during turns, improving steering efficiency and reducing stress on steering components. It's fundamental for smooth cornering.

steering geometry - toe-out on turn

Steering Ratio

$$R_s = \frac{\text{Steering Wheel Angle}}{\text{Road Wheel Angle}}$$

Symbol	Meaning
R_s	Steering Ratio
Steering Wheel Angle	Total rotation of the steering wheel
Road Wheel Angle	Average angle of the steered road wheels

Example

A car's steering wheel turns 3 full rotations (1080 degrees) from lock-to-lock. The road wheels turn 30 degrees from center to full lock. Calculate the steering ratio.

First, find the total road wheel angle from lock-to-lock: $30 \text{ degrees} \times 2 = 60 \text{ degrees}$.

Then, apply the formula: $R_s = \frac{1080 \text{ degrees}}{60 \text{ degrees}} = 18$.

The steering ratio is 18:1.

Applies when:

Typically measured from lock-to-lock or for a specific turn.